

LEDGER · 2026-09-01 · MATERIALS SCIENCE

Cement-Free Refractory Castables via Potassium-Activated Metakaolin Geopolymerization

PRODUCT



A product dossier of The Absence Audit. The current audit flags this concept — Margin fails the operating-cost check — and the full disclosure ships inside the dossier.

Incumbent benchmark

HarbisonWalker Mizzou® Castable Plus

Entry application

mid-temperature forge furnace linings and burner blocks

Measured advantage

Maximum service temperature of ~1250°C vs incumbent's 1650°C (0.75x delta, strictly inferior)

Time to first revenue	12 months
Gross margin at parity	49.6%
Startup CapEx	\$315,000
Payback from first sale	7.9 months
Capital productivity	4.24x
Regulatory risk	Low — no material regulatory CapEx reported (\$0 estimated)

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Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

Buy the full dossier — \$299.00 →

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

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